

EXTRAGALACTIC CATALOGS

CONTENTS

The catalogs must include galaxies *and* stars

MANDATORY COLUMNS FOR MOONLIGHT

(No NULL entries allowed)

- **ID:** A string uniquely identifying the source within the catalog (see below)
- **RA, DEC:** Right Ascension and Declination of the source, in decimal degrees, ICRS reference system.
- **PMRA, PMDEC:** proper motion at EPOCH 2016 (values for stars, 0.0 for galaxies)
- **MAG:** magnitude in the H_{AB} band for *all* sources - both extragalactic objects AND stars
- **SIZE:** source *diameter*, in arcsec. This should be a size indicator that *encircles* the majority of the source flux above the background (if not available = -99)

MANDATORY COLUMNS FOR SCIENTIFIC CLASSES IDENTIFICATION

(e.g. Passive, Star-forming, High-z, AGN, ... - No NULL entries allowed)

- **Z:** redshift estimate (photometric or spectroscopic; if not available = -99).
- **ABS_U, ABS_V, ABS_J:** U, V, J absolute magnitude (if not available or not derived (e.g. not from Bagpipes) = 99)
- **in_XXX_sample:** binary flags identifying membership in a given scientific sample (yes = 1; no = 0)

Examples include:

- `in_PaA`
- `in_AGN`
- `in_JWST`
- etc.

Scientific classes may overlap (e.g. `Passive && in_AGN == 1`).

The data-flow system will not attempt to resolve such overlaps, and the target will inherit the highest assigned priority and observing time among the matching classes.

Additional columns should be included whenever the origin of a quantity is not unique or unambiguous, in order to track its provenance (e.g. JWST spectroscopic and photometric redshift, source catalog, etc...).

OTHER MANDATORY COLUMNS FOR STARS

(No NULL entries allowed)

- **STAR (1,0)**: binary flags identifying if object is a star (yes = 1; no = 0)
 - **GAIA_magR: Johnson R-band magnitude in the Vega system** (for galaxies = -99)
 - **RUWE** (for galaxies = -99)
- **All the parameters necessary to identify P1 targets; – TBD**
- **Telluric & check stars; -TBD**

Objects flagged as stars (**STAR == 1**) **must not overlap** with any extragalactic scientific class (e.g. no object with **STAR == 1 && in_AGN == 1**).

ANCILLARY DATA (up to SWGs)

This catalog will also serve as the *master science parent catalog*.

Any additional information potentially useful for the SWGs may therefore be included, such as:

- **GAIA STAR_ID**
- **PHOTO_Z_ERROR**
- **PHOTO_Z_ERROR**
- **PHOTOMETRIC DATA** used to derive photometric redshifts and absolute magnitudes (99 if unavailable)
- **ABS_MAGS** available in the original catalogue (not available = 99)
- **MASS and other physical parameters** if available from SED fitting (not available = -99) [not currently used to define classes, but it could be useful for a-posteriori cross checks]
- **other SIZES** (not available = -99)
- **other physical parameters** available from SED fitting (not available value depends on each parameter range)

SOME RECIPES

• GALAXIES

For galaxies without an available H-band magnitude, a placeholder value **must be assigned** in the MAG column:

- 99 → source expected to be fainter than the magnitude limit (ML) of the background-source sample
- 98 → source expected to be brighter than the magnitude limit (ML) of the background-source sample
- 97 → very bright source ($H < 12$)

• STARS

- Primary selection from Gaia → Gaia_star_ID >0 → available G and R
- Cross-match with 2MASS
- For those not in 2MASS cross-match with near-IR catalogues
- If not in any near-IR catalogue → extrapolate mag-H from Gaia colours (accept larger errors)
- **ALL STARS MUST have a mag_H**

• ID

|**XX**|0|**YY**|**ZZZZZZZZZZ**|

XX : field

COSMOS	11
CDFS	12
UDS	13
XMM	14

0 : fixed to create some space

YY : catalog origin

H band selected COSMOS catalog	01
I band selected additional catalog	02
High-Z	03
AGN	04
Lega-C	05

ZZZZZZZZZZ : 9-digits for the sequential number

Examples:

- A Lega-C galaxy in COSMOS: 11005000000327
- An AGN in CDFS: 12004000000231

NOTES

- **Source IDs must remain stable throughout the entire GTO period.** Updated versions of the catalogs are allowed, but previously existing sources must retain their original IDs.
- If a source is removed from a catalog, its ID must never be reassigned to a different object.
- New sources may be added in subsequent catalog releases, provided that they are assigned new unique IDs.
- Catalog providers are responsible for checking duplicated sources, as well as ensuring the consistency and accuracy of the astrometry at ANY implementation of the catalog.